

GROUP BY with JOIN?

Consider two tables.

Customers

customer_id	customer_name
1	Arpit
2	Aadz
3	Rahul

Orders

order_id	customer_id	amount
101	1	1200
102	1	800
103	2	1500
104	1	600
105	3	900

Question

How many orders has each customer placed?

Customer names are in one table.

Orders are in another.

Need a JOIN first.

14.11.2 Basic Pattern

```
SELECT group_column,  
       AGG(...)  
FROM TableA  
JOIN TableB  
ON ...  
GROUP BY group_column;
```

Remember:

JOIN first, GROUP later. Example — Orders per Customer

```
SELECT c.customer_name,  
       COUNT(*) AS total_orders  
FROM Customers c  
JOIN Orders o  
ON c.customer_id = o.customer_id  
GROUP BY c.customer_name;
```

Total Spending per Customer

```
SELECT c.customer_name,  
       SUM(o.amount) AS total_spent  
FROM Customers c  
JOIN Orders o  
ON c.customer_id = o.customer_id  
GROUP BY c.customer_name;
```

Movies per Director

Directors

id	director
1	Christopher Nolan
2	Rajkumar Hirani

Movies

movie	director_id
Interstellar	1
Inception	1
3 Idiots	2

Query

```
SELECT d.director,  
       COUNT(*) AS movies  
FROM Directors d  
JOIN Movies m  
ON d.id=m.director_id  
GROUP BY d.director;
```

LEFT JOIN + GROUP BY

Interview favorite.

Question

Show all customers, including those who never ordered.

```
SELECT c.customer_name,  
       COUNT(o.order_id) AS orders  
FROM Customers c  
LEFT JOIN Orders o  
ON c.customer_id=o.customer_id  
GROUP BY c.customer_name;
```

COUNT(o.order_id) ignores NULL values. Suppose we have:

Students

Student	Gender	Result
Arpit	Male	Pass
Aadz	Female	Pass
Rahul	Male	Fail
Priya	Female	Pass
Riya	Female	Fail

Question

Find

Male students

Female students

```
SELECT
SUM(CASE
    WHEN gender='Male'
    THEN 1
    ELSE 0
    END) AS male_count,

SUM(CASE
    WHEN gender='Female'
    THEN 1
    ELSE 0
    END) AS female_count
FROM Students;
```

Conditional AVG

Average salary of ECE employees.

```
SELECT
AVG(CASE
    WHEN department='ECE'
    THEN salary
    END
) AS avg_ece_salary
FROM Employee;
```

Conditional Aggregation Patterns

★★★★★ Master these patterns—they solve dozens of interview questions.

Think of this as a toolbox.

Pattern 1 — Count Matching Rows

```
SUM(CASE
    WHEN condition
    THEN 1
    ELSE 0
END)
```

Example

Count delivered orders.

```
SELECT
SUM(CASE
    WHEN status='Delivered'
    THEN 1
    ELSE 0
END)
FROM Orders;
```

Pattern 2 — Sum Matching Values

```
SUM(CASE
    WHEN condition
    THEN amount
    ELSE 0
END)
```

Example

Delivered revenue.

Pattern 3 — Average Matching Values

```
AVG(CASE
    WHEN condition
    THEN salary
END)
```

Example

Average salary of ECE employees.

Pattern 4 — Multiple Counts in One Query

```
SELECT

SUM(CASE WHEN gender='Male' THEN 1 ELSE 0 END) male,

SUM(CASE WHEN gender='Female' THEN 1 ELSE 0 END) female
FROM Students;
```

Pattern 5 — Percentage

Interview favorite.

Percentage of delivered orders.

```
SELECT
100.0 *

SUM(CASE
    WHEN status='Delivered'
    THEN 1
    ELSE 0
END)
/COUNT(*) AS delivery_percentage
FROM Orders;
```

Pattern 6 — Ratio

Male-to-female ratio.

```
SELECT

SUM(CASE WHEN gender='Male' THEN 1 ELSE 0 END)/

SUM(CASE WHEN gender='Female' THEN 1 ELSE 0 END)
AS ratio
FROM Students;
```

(Guard against division by zero in production code.)

Pattern 7 — Pivot Simulation

Convert rows into columns.

Orders

Status
Delivered
Pending
Cancelled

Query

```
SELECT

SUM(CASE WHEN status='Delivered' THEN 1 ELSE 0 END) AS delivered,

SUM(CASE WHEN status='Pending' THEN 1 ELSE 0 END) AS pending,

SUM(CASE WHEN status='Cancelled' THEN 1 ELSE 0 END) AS cancelled
FROM Orders;
```

Output

Delivered	Pending	Cancelled
15	6	2